ST2334: Probability and Distributions

1. Events and counting

S: possible outcomes; an event $A \subseteq S$; A^c: complement; $A \cap B$: both; $A \cup B$: either.

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

$P(S) = 1$, $P(\varnothing) = 0$, $P(A^c) = 1 - P(A)$. Relative frequency n_A/n approaches $P(A)$ as trials grow.

Counting: For r distinct items selected from n: order matters $\Rightarrow {}_nP_r = n!/(n-r)!$; order does not matter $\Rightarrow \binom{n}{r} = n!/[r!(n-r)!]$.

Counting reminder: These formulas choose without repetition; ${}_nP_r = r!\binom{n}{r}$.

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$; if disjoint, $P(A \cap B) = 0$. Equally likely outcomes: $P(A) = |A|/|S|$.

Inclusion–exclusion: For overlapping events: add singles, subtract pairs, add triples, and alternate. No independence assumption is needed.

Three events: $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.

In general, $P(\bigcup_{i=1}^n A_i) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \le i_1 < \cdots < i_k \le n} P(A_{i_1} \cap \cdots \cap A_{i_k})$.

Birthday bound: For k independent items in m equally likely slots, $P(\text{collision}) = 1 - \prod_{i=0}^{k-1}(1 - i/m)$; its 50% point is near $1.18\sqrt{m}$.

Union bound: $P(\bigcup_i A_i) \le \sum_i P(A_i)$, without independence. For n independent assignments to m slots, collision probability $\le \binom{n}{2}/m$.

2. Conditional probability

Use: Update a probability after learning A; require $P(A) > 0$. For an "at least one" event, try $1 - P(\text{none})$.

$P(B \mid A) = P(A \cap B)/P(A)$, hence

$$P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B).$$

Chain: $P(A \cap B \cap C) = P(A)P(B \mid A)P(C \mid A \cap B)$.

Independence: $A \perp B \iff P(A \cap B) = P(A)P(B)$. Then $P(B \mid A) = P(B)$ when $P(A) > 0$. Positive-probability disjoint events are dependent. Complements of independent events remain independent.

Use: Split a target event B into exhaustive, disjoint cases A_i.

$$P(B) = \sum_i P(A_i)P(B \mid A_i).$$

Two cases: $P(B) = P(A)P(B \mid A) + P(A^c)P(B \mid A^c)$. **Use:** Reverse evidence-to-cause probabilities with Bayes' rule.

$$P(A_k \mid B) = \frac{P(A_k)P(B \mid A_k)}{\sum_i P(A_i)P(B \mid A_i)}.$$

Two-case Bayes: For cases A, A^c, use $P(A \mid B) = P(A)P(B \mid A)/P(B)$ with the two-case $P(B)$ above.

Conditional product: Always start with $P(A \cap B \cap C) = P(A)P(B \mid A)P(C \mid A \cap B)$. If $B \perp C \mid A$, replace the last term by $P(C \mid A)$; count $P(A)$ only once.

Simpson's paradox: An aggregate comparison can reverse every subgroup comparison when subgroup weights differ; inspect conditional rates.

3. One random variable

$X : S \to \mathbb{R}$ assigns a number to each outcome; R_X is its support/range. Discrete: countable values. Continuous: density over intervals.

PMF, discrete: $f_X(x) = P(X = x) \ge 0$, $\sum_{x \in R_X} f_X(x) = 1$. Add masses to find $P(X \in B) = \sum_{x \in B} f_X(x)$.

PDF, continuous: $f_X(x) \ge 0$, $\int f_X(x)\,dx = 1$. Integrate area: $P(a \le X \le b) = \int_a^b f_X(x)\,dx$. Single points have probability zero; density height is not probability and may exceed 1.

CDF, either type: $F_X(x) = P(X \le x)$. Discrete: jump at x is $P(X = x)$, and $P(a \le X \le b) = F_X(b) - F_X(a^-)$. Continuous: $F_X(x) = \int_{-\infty}^x f_X(t)dt$, $f_X = F_X'$, and interval probability is $F_X(b) - F_X(a)$. Invert F_X for a quantile.

CDF check: F_X is right-continuous, never decreases, and tends to 0/1 at the extremes; inversion turns a percentile into a value.

Use: Find an average or a transformed average without first finding the distribution of $g(X)$.

$$E[g(X)] = \sum_x g(x)f_X(x) \quad \text{or} \quad \int g(x)f_X(x)dx.$$

$\mu_X = E(X)$, $E(aX + bY + c) = aE(X) + bE(Y) + c$, even without independence.

$$\text{Var}(X) = E[(X - \mu_X)^2] = E(X^2) - \mu_X^2.$$

$\text{Var}(aX + b) = a^2 \text{Var}(X)$. $\sigma_X = \sqrt{V(X)}$ has the units of X; variance has squared units.

Tail-sum expectation: For nonnegative integer X, $E(X) = \sum_{k \ge 1} P(X \ge k) = \sum_{k \ge 0} P(X > k)$. For nonnegative continuous T, $E(T) = \int_0^\infty P(T > t)\,dt$. Use when the tail is easier than the PMF/PDF.

Inspection paradox: A random arrival is more likely to fall in a long gap, so the expected wait can exceed what a naive average-gap calculation suggests.

4. Joint distributions

$f_{X,Y}(x, y)$: joint mass/density on the allowed pairs. For a region D, sum masses or integrate $f_{X,Y}$ over D to get $P((X, Y) \in D)$. To ignore Y, marginalize:

$$f_X(x) = \sum_y f_{X,Y}(x, y) \quad \text{or} \quad \int f_{X,Y}(x, y)\,dy.$$

Swap X, Y to get $f_Y(y)$. For a joint function g, $E[g(X, Y)] = \sum_{x,y} g(x, y)f_{X,Y}(x, y)$ or a double integral. **Use:** Fix $X = x$ and renormalize its slice; require $f_X(x) > 0$.

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}, \qquad f_{X,Y} = f_X f_{Y|X}.$$

Conditional mean (regression function): $E(Y \mid X = x) = \sum_y y f_{Y|X}(y \mid x)$, or integrate in y. It predicts Y from $X = x$ under squared error.

Independent variables: $X \perp Y \iff f_{X,Y}(x, y) = f_X(x)f_Y(y)$. Quick test: support is a product with no forbidden pairs, and $f_{X,Y} = C\,g(x)h(y)$; g, h need not be PMFs/PDFs. Then $E[g_1(X)h_1(Y)] = E[g_1(X)]E[h_1(Y)]$.

Use: Measure joint linear movement or the variance of a sum.

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - E(X)E(Y),$$

$$\text{Var}(aX + bY) = a^2V(X) + b^2V(Y) + 2ab\,\text{Cov}(X, Y).$$

$\text{Cov}(aX + b, cY + d) = ac\,\text{Cov}(X, Y)$. Independence gives covariance zero, but zero covariance need not mean independence.

Independent difference: If $X \perp Y$, $V(X + Y) = V(X - Y) = V(X) + V(Y)$; the minus sign does not subtract variance.

For many variables: $V(\sum_i X_i) = \sum_i V(X_i) + 2\sum_{i<j} \text{Cov}(X_i, X_j)$. $\rho_{X,Y} = \text{Cov}(X, Y)/(\sigma_X\sigma_Y)$, where $\sigma_X = \sqrt{V(X)}$; $-1 \le \rho \le 1$. Correlation is unit-free and measures linear association.

5. Discrete families

Let $p = P(\text{success})$, $q = 1 - p = P(\text{failure})$; n is fixed trials; k is a target count; x is the realized value. Bernoulli process: identical, independent trials.

Which count?

Uniform: k equiprobable values x_i: $P(X = x_i) = 1/k$, $E(X) = \sum x_i/k$, $V(X) = \sum x_i^2/k - [E(X)]^2$.

Bernoulli: One success/failure trial: $X \sim \text{Bern}(p)$, $P(X = x) = p^x q^{1-x}$, $x = 0, 1$; $E = p$, $V = pq$.

Binomial: Successes in n trials: $X \sim \text{Bin}(n, p)$, $P(X = x) = \binom{n}{x} p^x q^{n-x}$, $x = 0, \ldots, n$; $E = np$, $V = npq$.

Geometric: Trials through first success ($k = 1$ in NB): $P(X = x) = q^{x-1}p$, $x \ge 1$; $E = 1/p$, $V = q/p^2$.

Negative binomial: Trials until kth success: $P(X = x) = \binom{x-1}{k-1} p^k q^{x-k}$, $x \ge k$; $E = k/p$, $V = kq/p^2$.

Poisson: Count in a fixed window: $P(X = x) = e^{-\lambda}\lambda^x/x!$, $x \ge 0$; $E = V = \lambda$. For rate r per unit and window t, $\lambda = rt$; disjoint-window counts are independent in a Poisson process.

Rare binomial approximation: For large n, small p, fixed $\lambda = np$: $\text{Bin}(n, p) \approx \text{Pois}(\lambda)$.

Benford: For the leading digit $D \in \{1, \ldots, 9\}$ of some scale-spanning data, $P(D = d) = \log_{10}(1 + 1/d)$; it is not a universal law for every dataset.

6. Continuous families

Choose by the question

Uniform $U(a, b)$**:** Equal density over $[a, b]$: $f(x) = 1/(b - a)$; $F(x) = (x - a)/(b - a)$ for $a \le x \le b$; $E = (a + b)/2$, $V = (b - a)^2/12$. For $U \sim U(0, 1)$, $P(U \le u) = u$, $0 \le u \le 1$.

Exponential $\text{Exp}(\lambda)$**:** Wait for the next event in a Poisson process with rate $\lambda > 0$: $f(x) = \lambda e^{-\lambda x}$, $F(x) = 1 - e^{-\lambda x}$, $P(X > x) = e^{-\lambda x}$ for $x \ge 0$; $E = 1/\lambda$, $V = 1/\lambda^2$.

Normal $N(\mu, \sigma^2)$**:** Symmetric bell shape, often for sums of many small independent effects: $f(x) = e^{-(x-\mu)^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$; $E = \mu$, $V = \sigma^2$.

Exponential memorylessness: $P(X > s + t \mid X > s) = P(X > t)$, $s, t \ge 0$. λ is events per unit time; its reciprocal is the mean wait.

Pareto reset: For $T \sim \text{Pareto}(x_m, \alpha)$, $P(T > t) = (x_m/t)^\alpha$, $t \ge x_m$. It is not memoryless. If $\alpha > 1$ and restarting costs nothing, a fresh run has smaller expected wait than continuing after $t > \alpha x_m$; otherwise compare costs.

Rate reminder: For exponential waits use the rate, not a count over an arbitrary window; if mean wait is μ, set $\lambda = 1/\mu$.

Uniform annulus: For a point uniform in area with radius $2 \le R \le 5$, $F_R(r) = (r^2 - 4)/21$ for $2 \le r \le 5$. To sample it, take $U \sim U(0, 1)$ and set $R = \sqrt{4 + 21U}$; start with $P(R \le r)$.

Use: Convert a normal value to standard normal Z and read $\Phi(z) = P(Z \le z)$.

$$Z = (X - \mu)/\sigma \sim N(0, 1), \qquad P(X \le x) = \Phi((x - \mu)/\sigma).$$

Normal symmetry: $\Phi(z) = 1 - \Phi(-z)$. Upper-tail cutoff z_α: $P(Z \ge z_\alpha) = \alpha$; $z_{0.05} \approx 1.645$, $z_{0.025} \approx 1.960$, $z_{0.01} \approx 2.326$.

Binomial to normal: If $np > 5$ and $nq > 5$ (guideline), approximate $X \sim \text{Bin}(n, p)$ by $Y \sim N(np, npq)$. Apply a 0.5 continuity correction: $P(X \le k) \approx P(Y < k + 0.5)$, $P(X = k) \approx P(k - 0.5 < Y < k + 0.5)$, and $P(X > k) \approx P(Y > k + 0.5)$.